Chapter 1, Problem 1P

Problem

How much charge is represented by these number of electrons?

(a) 6.482 × 1017

(b) 1.24 × 1018

(c) 2.46 × 1019

(d) 1.628 × 1020

Step-by-step solution

Step 1 of 4

(a)

The number of electrons is 6.48×10^{17}.

Multiply the number of electrons with the charge of the electron, e to obtain the total charge.

Calculate the number of coulombs represented by 6.48×10^{17} electrons.

$$\begin{aligned} Q &= (\text{number of electrons})\times(e) \\ &= (6.482\times10^{17})\times(1.602\times10^{-19}) \\ &= 0.10384\ \text{C} \end{aligned}$$

Therefore, the number of coulombs, Q represented by 6.48×10^{17} electrons are, $\boxed{0.10384\ \text{C}}$

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Step 2 of 4

(b)

The number of electrons is 1.24×10^{18}.

Multiply the number of electrons with the charge of the electron, e to obtain the total charge.

Calculate the number of coulombs represented by 1.24×10^{18} electrons.

$$\begin{aligned} Q &= (\text{number of electrons})\times(e) \\ &= (1.24\times10^{18})\times(1.602\times10^{-19}) \\ &= 0.19864\ \text{C} \end{aligned}$$

Therefore, the number of coulombs, Q represented by 1.24×10^{18} electrons are, $\boxed{0.19864\ \text{C}}$.

Step 3 of 4

(c)

The number of electrons is 2.46×10^{19}

Multiply the number of electrons with the charge of the electron, e to obtain the total charge.

Calculate the number of coulombs represented by 2.46×10^{19} electrons.

$$\begin{aligned} Q &= (\text{number of electrons})\times(e) \\ &= (2.46\times10^{19})\times(1.602\times10^{-19}) \\ &= 3.9409\ \text{C} \end{aligned}$$

Therefore, the number of coulombs, Q represented by 2.46×10^{19} electrons are $\boxed{3.9409\ \text{C}}$.

Step 4 of 4

(d)

The number of electrons is 1.628×10^{20}.

Multiply the number of electrons with the charge of the electron, e to obtain the total charge.

Calculate the number of coulombs represented by 1.628×10^{20} electrons.

$$\begin{aligned} Q &= (\text{number of electrons})\times(e) \\ &= (1.628\times10^{20})\times(1.602\times10^{-19}) \\ &= 26.08\ \text{C} \end{aligned}$$

Therefore, the number of coulombs, Q represented by 1.628×10^{20} electrons are,

$\boxed{26.08\ \text{C}}$.